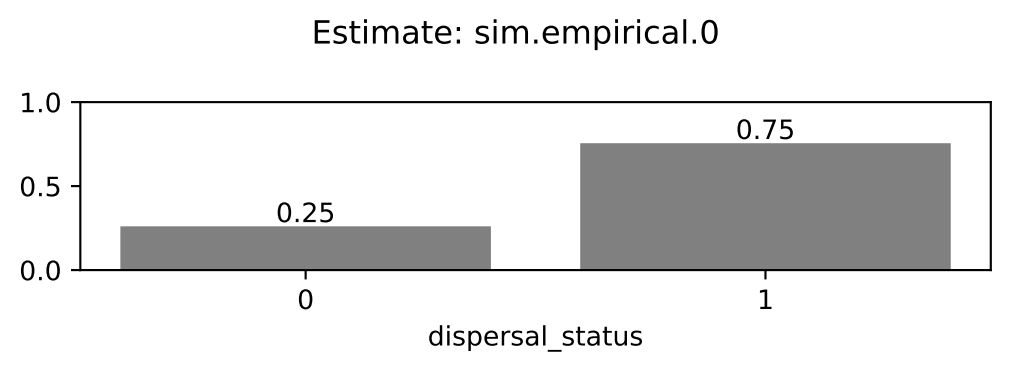
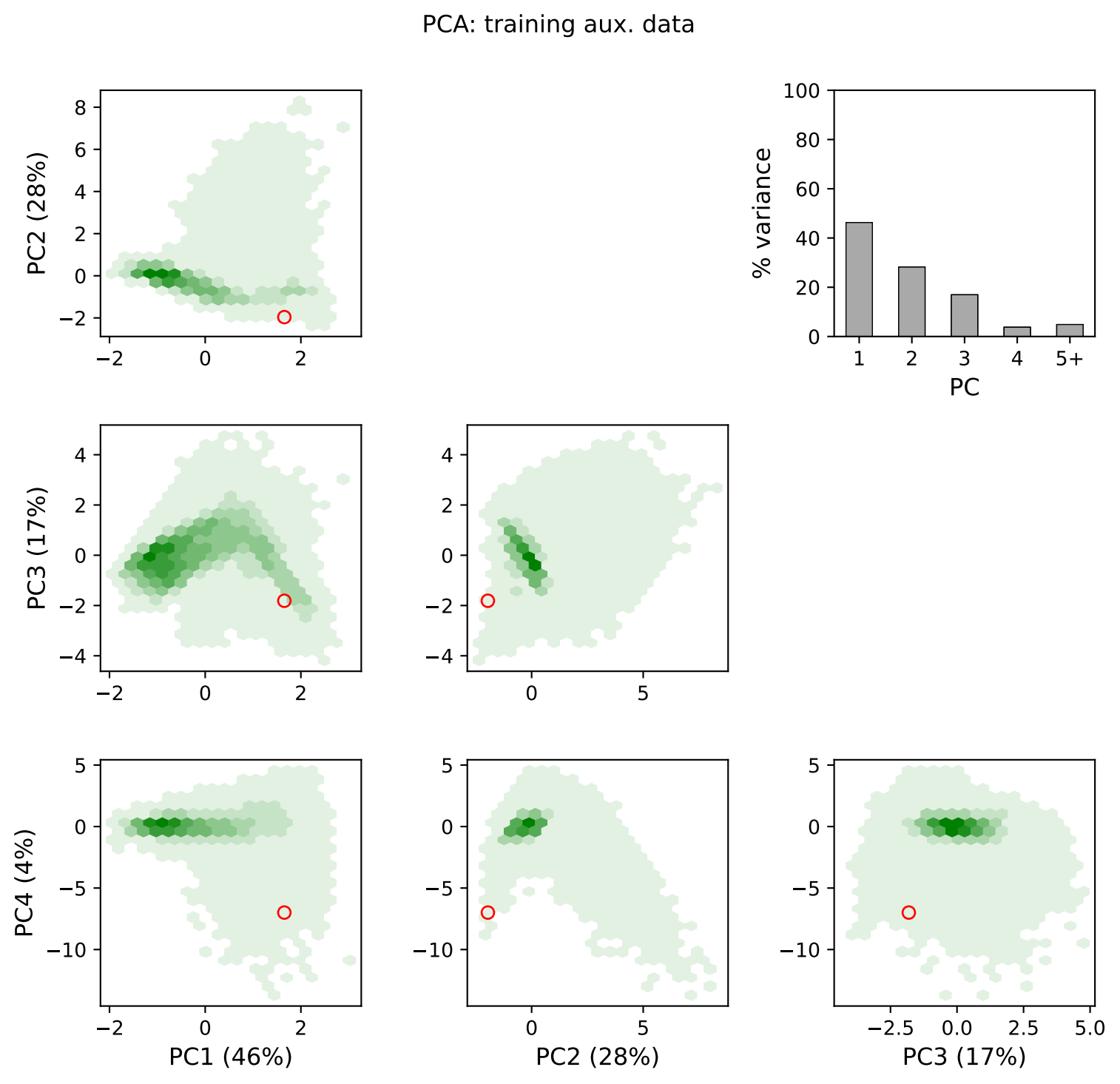
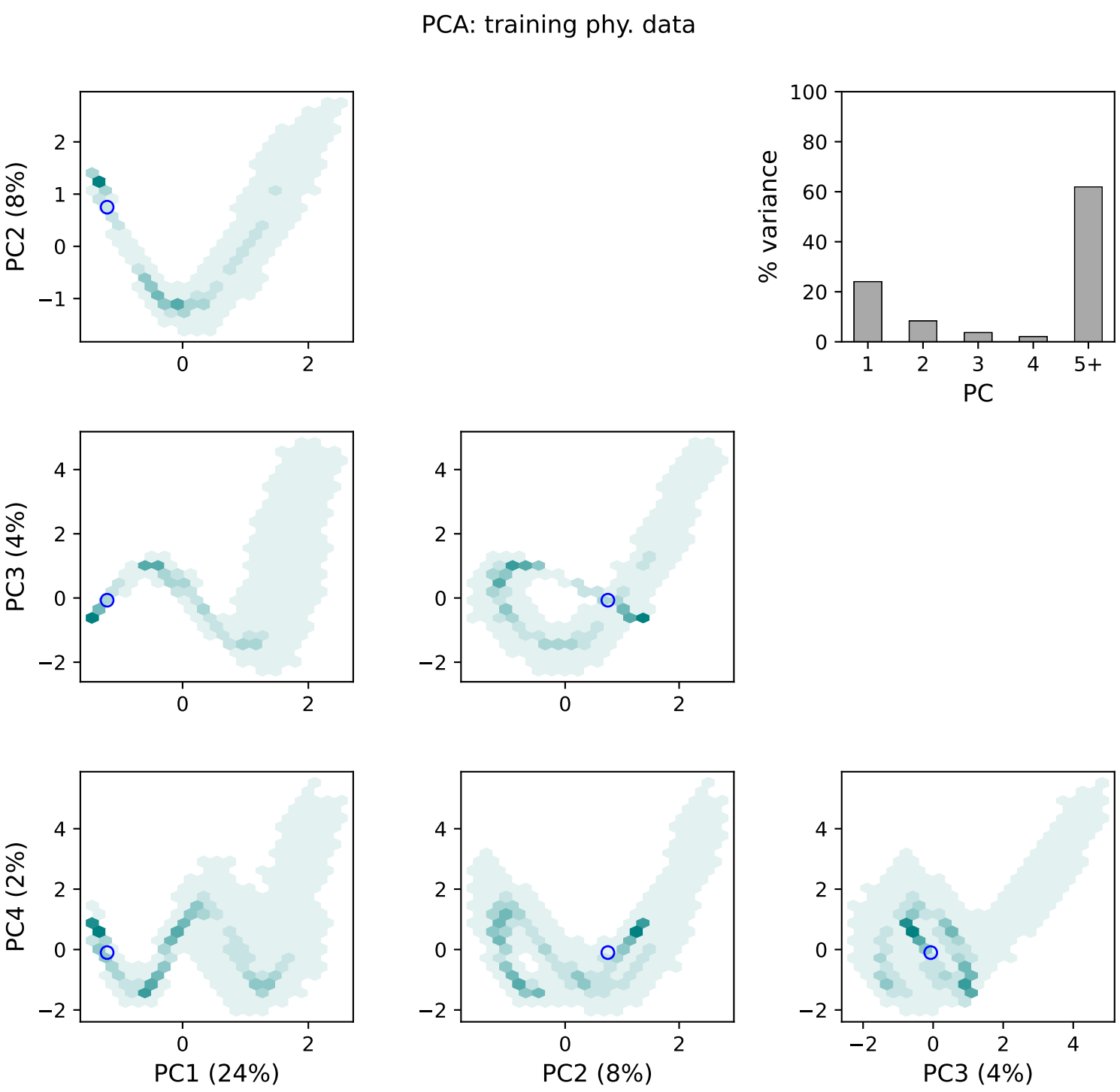
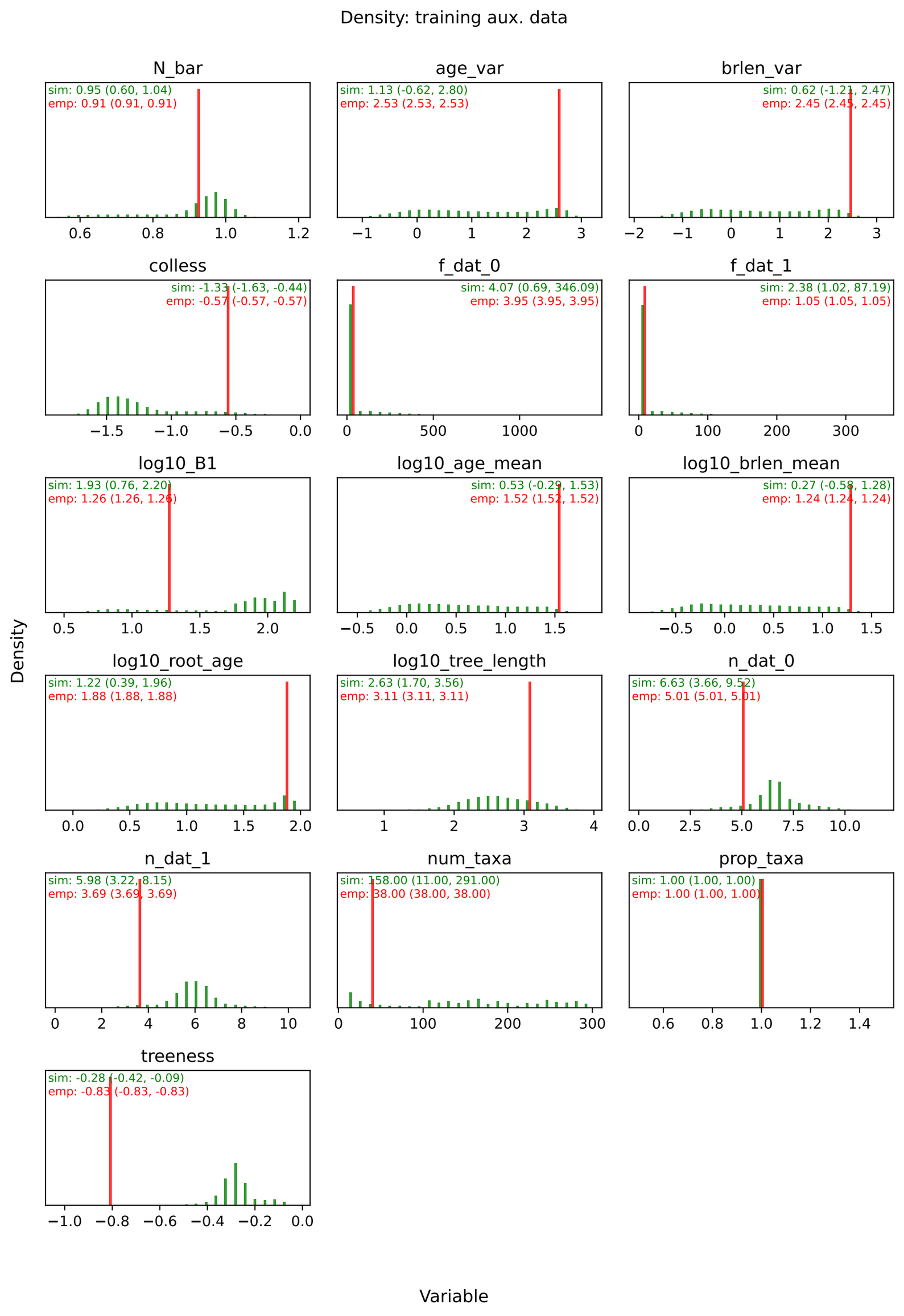
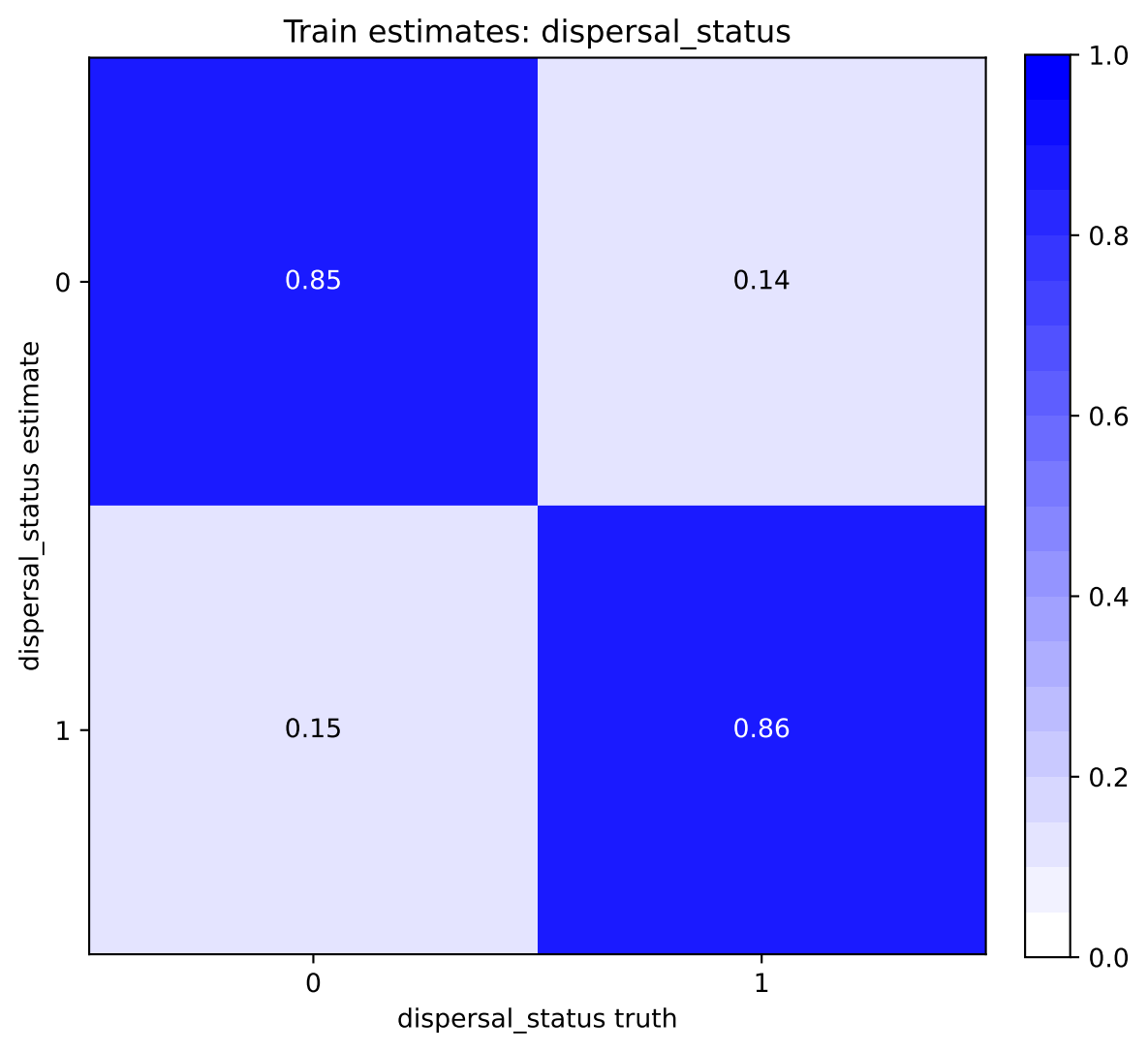
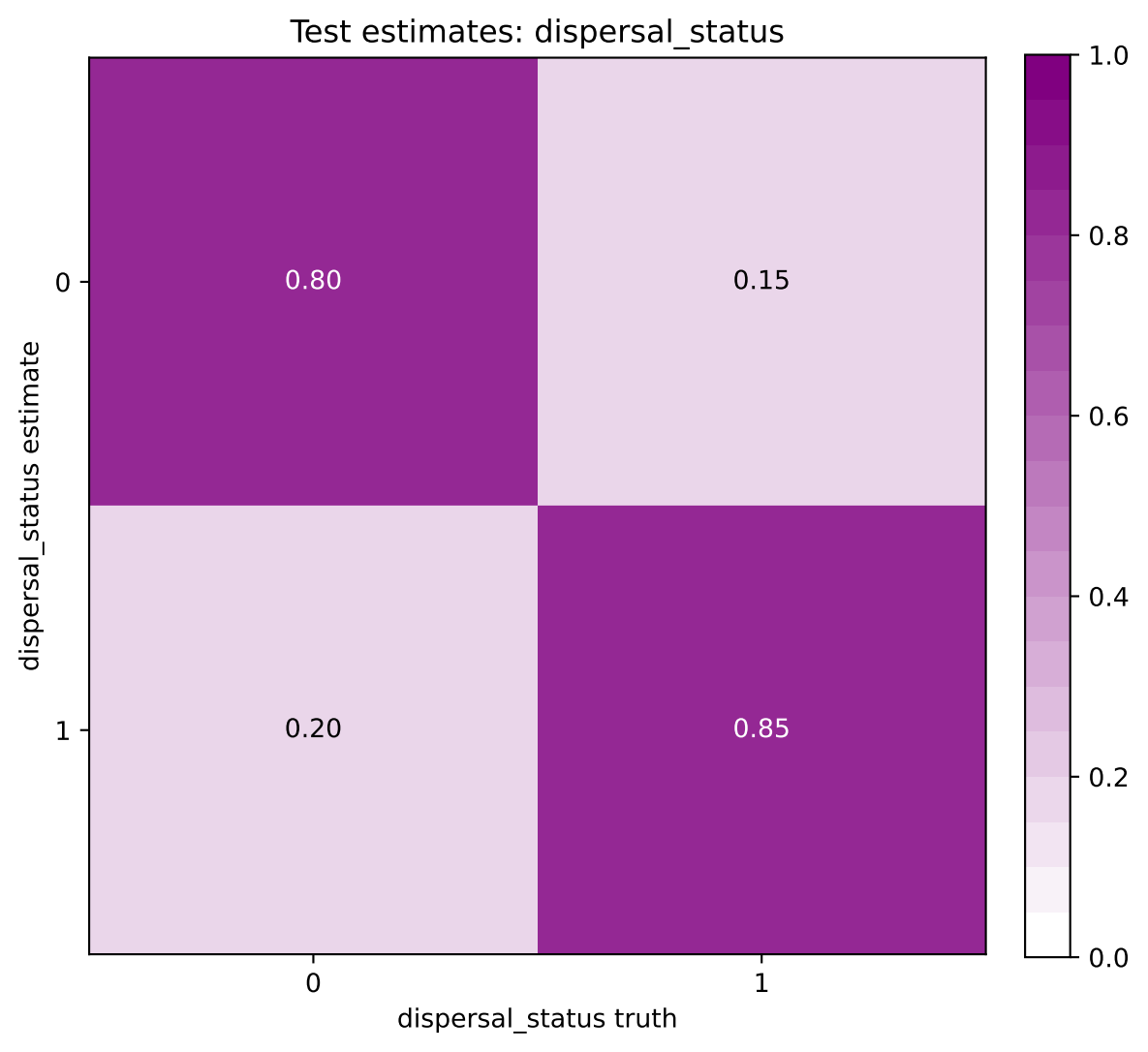

Estimate: sim.empirical.0
1.0
0.5
0.0
0.25
0.75
0
1
dispersal_status

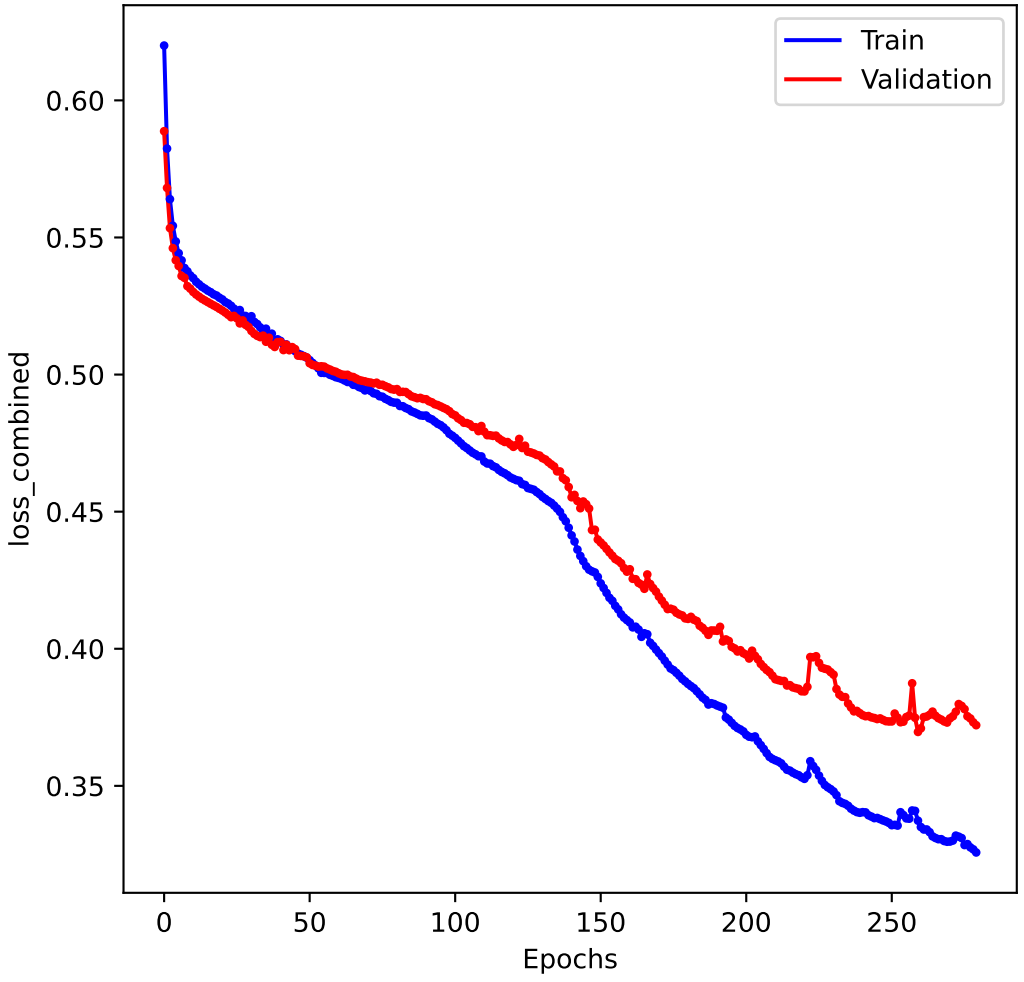

PCA: training aux. data
PC2 (28%)
% variance
PC
PC3 (17%)
PC4 (4%)
PC1 (46%)
PC2 (28%)
PC3 (17%)

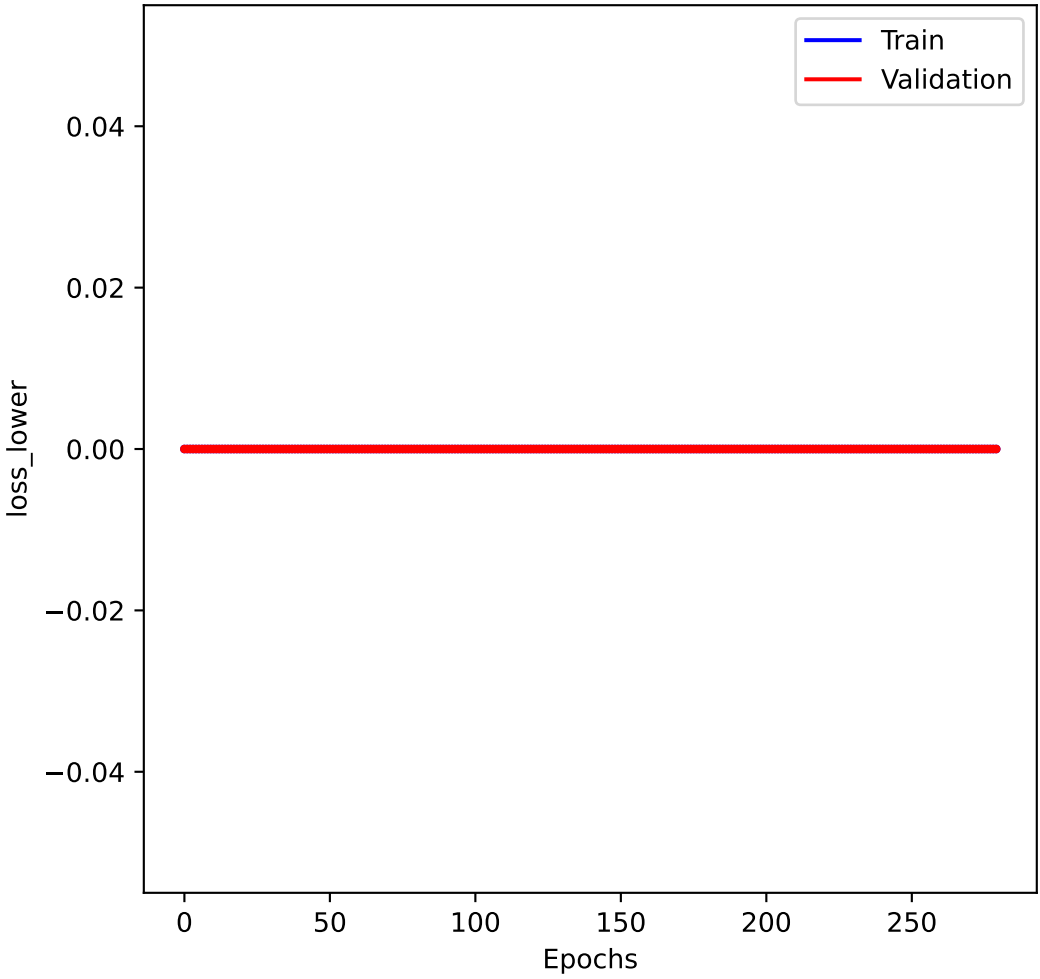

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (24%)
PC2 (8%)
PC3 (4%)

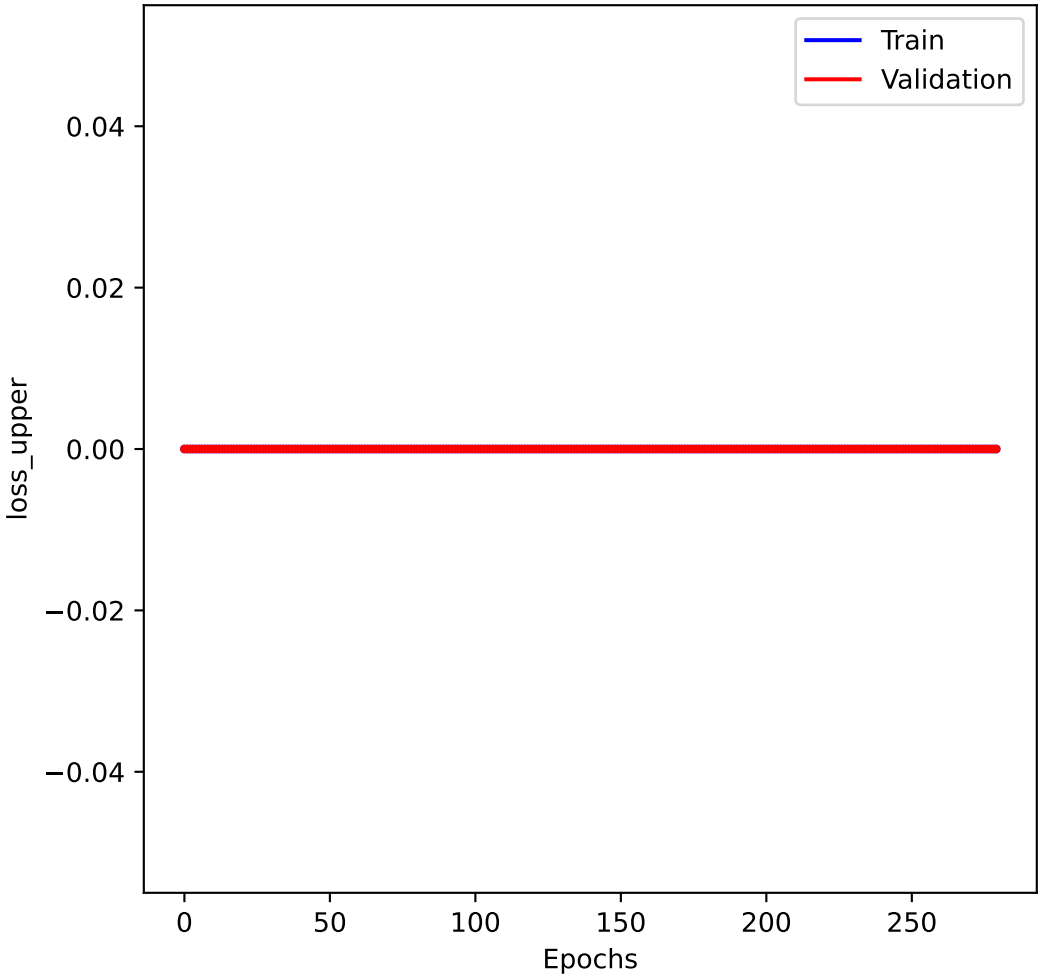

Density: training aux. data
N_bar
sim: 0.95 (0.60, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 1.13 (-0.62, 2.80)
emp: 2.53 (2.53, 2.53)
brlen_var
sim: 0.62 (-1.21, 2.47)
emp: 2.45 (2.45, 2.45)
colless
sim: -1.33 (-1.63, -0.44)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 4.07 (0.69, 346.09)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.38 (1.02, 87.19)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 1.93 (0.76, 2.20)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.53 (-0.29, 1.53)
emp: 1.52 (1.52, 1.52)
log10_brlen_mean
sim: 0.27 (-0.58, 1.28)
emp: 1.24 (1.24, 1.24)
log10_root_age
sim: 1.22 (0.39, 1.96)
emp: 1.88 (1.88, 1.88)
log10_tree_length
sim: 2.63 (1.70, 3.56)
emp: 3.11 (3.11, 3.11)
n_dat_0
sim: 6.63 (3.66, 9.52)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 5.98 (3.22, 8.15)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 158.00 (11.00, 291.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.42, -0.09)
emp: -0.83 (-0.83, -0.83)
Density
Variable

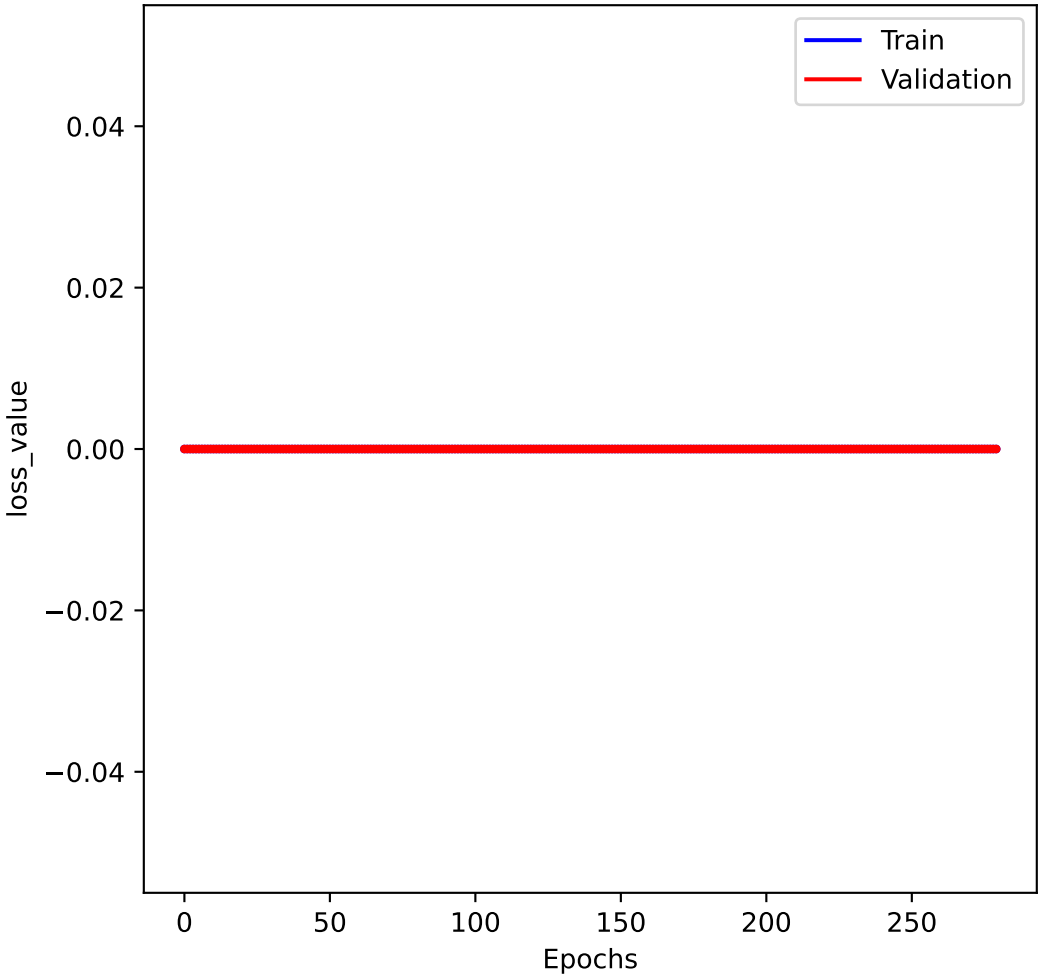

Train estimates: dispersal_status
0.85
0.14
0.15
0.86
0
1
dispersal_status estimate
0
1
dispersal_status truth
1.0
0.8
0.6
0.4
0.2
0.0

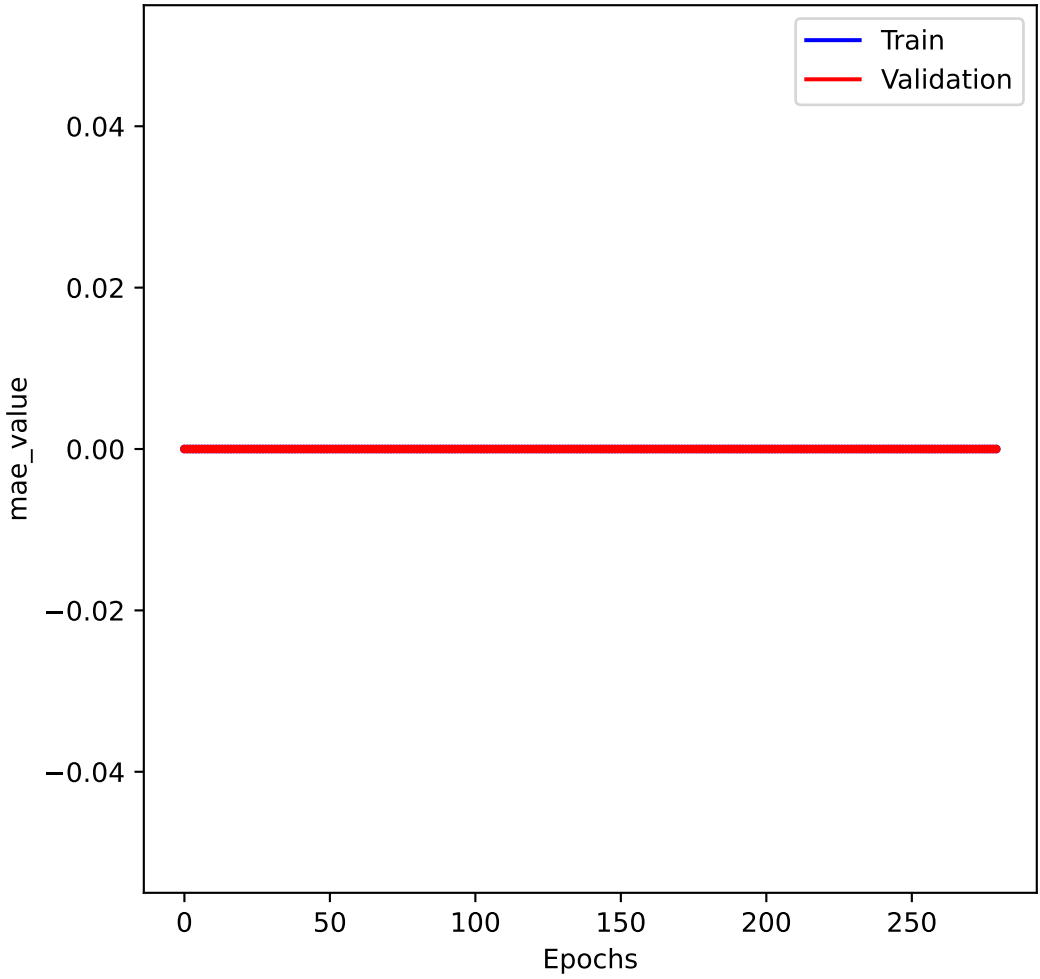

Test estimates: dispersal_status
0.80
0.15
0.20
0.85
0
1
dispersal_status estimate
0
1
dispersal_status truth
1.0
0.8
0.6
0.4
0.2
0.0

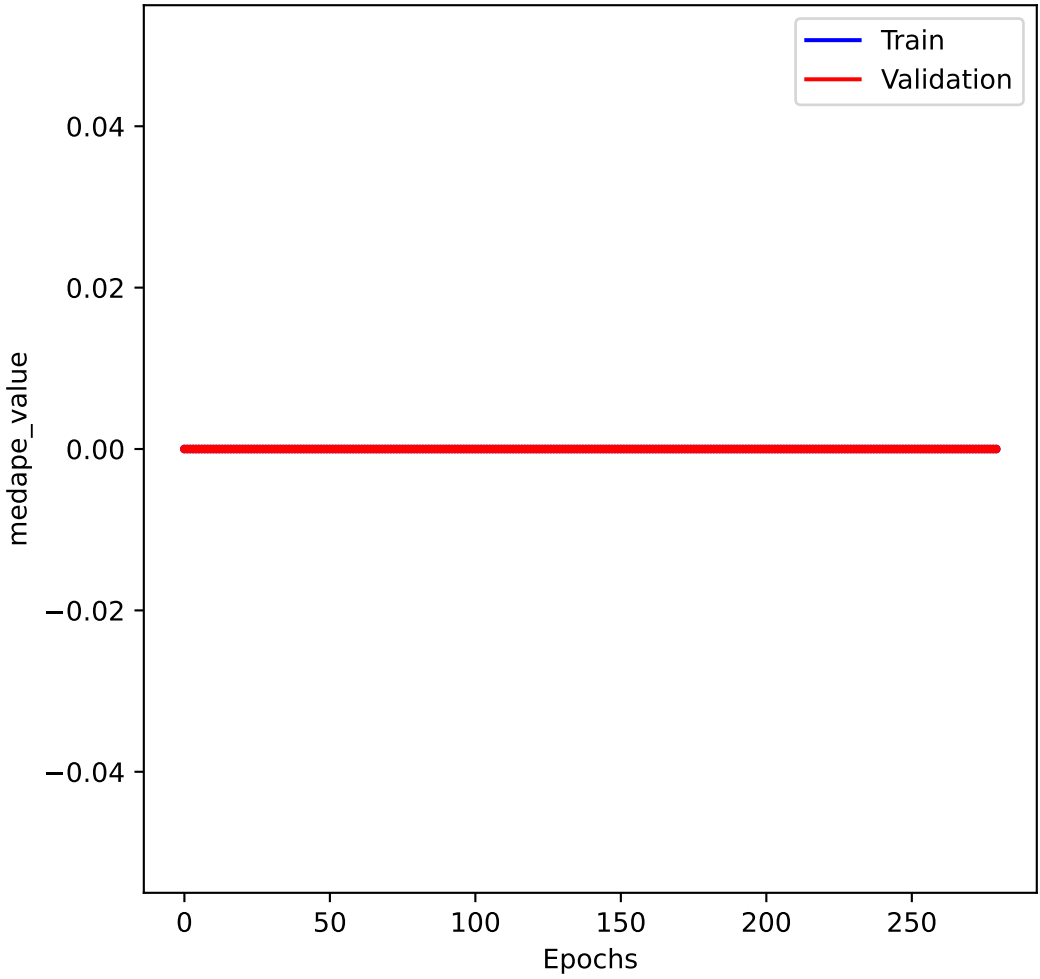

Train
Validation
loss_combined
0.60
0.55
0.50
0.45
0.40
0.35
0
50
100
150
200
250
Epochs

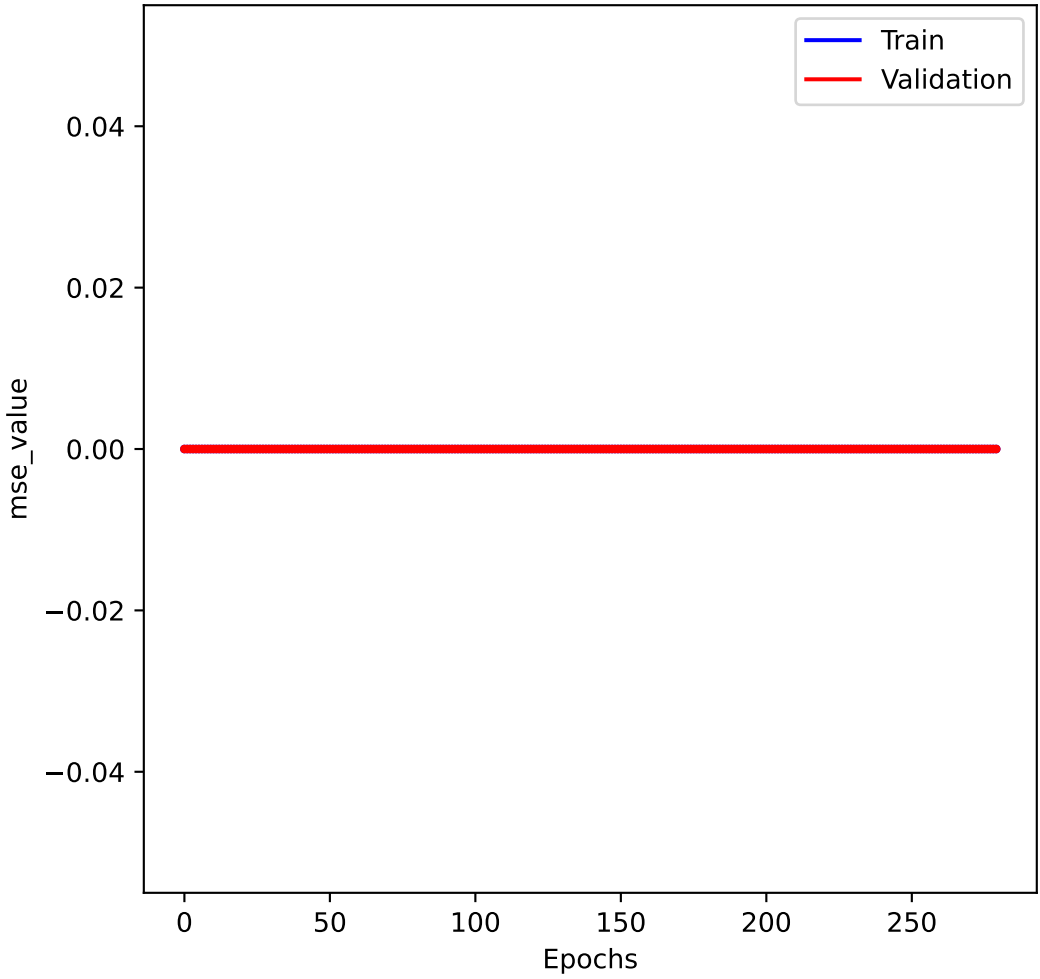

Train
Validation
0.04
0.02
0.00
−0.02
−0.04
loss_lower
0
50
100
150
200
250
Epochs

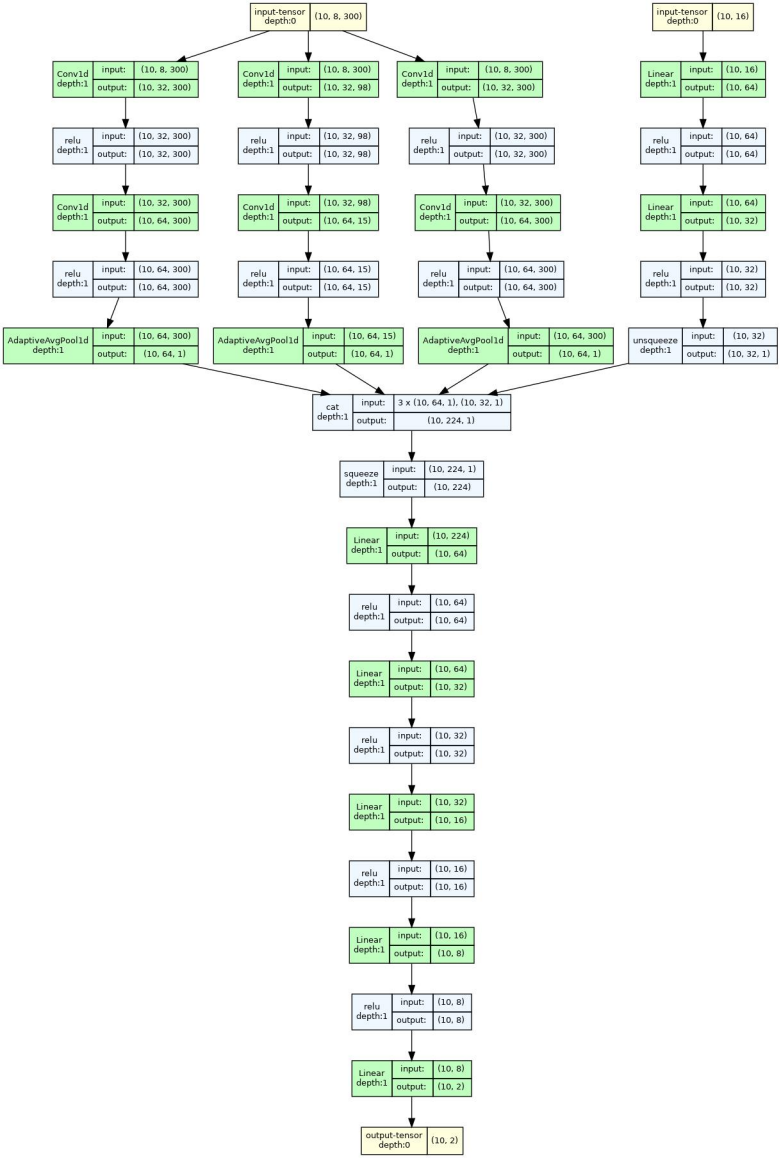

Train
Validation
0.04
0.02
0.00
−0.02
−0.04
loss_upper
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
loss_value
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
mae_value
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
medape_value
0
50
100
150
200
250
Epochs


Train
Validation
0.04
0.02
0.00
−0.02
−0.04
mse_value
0
50
100
150
200
250
Epochs


input-tensor depth:0 (10, 8, 300)
input-tensor depth:0 (10, 16)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 300)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 98)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 300)
Linear depth:1 input: (10, 16) output: (10, 64)
relu depth:1 input: (10, 32, 300) output: (10, 32, 300)
relu depth:1 input: (10, 32, 98) output: (10, 32, 98)
relu depth:1 input: (10, 32, 300) output: (10, 32, 300)
relu depth:1 input: (10, 64) output: (10, 64)
Conv1d depth:1 input: (10, 32, 300) output: (10, 64, 300)
Conv1d depth:1 input: (10, 32, 98) output: (10, 64, 15)
Conv1d depth:1 input: (10, 32, 300) output: (10, 64, 300)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64, 15) output: (10, 64, 15)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 32) output: (10, 32)
AdaptiveAvgPool1d depth:1 input: (10, 64, 300) output: (10, 64, 1)
AdaptiveAvgPool1d depth:1 input: (10, 64, 15) output: (10, 64, 1)
AdaptiveAvgPool1d depth:1 input: (10, 64, 300) output: (10, 64, 1)
unsqueeze depth:1 input: (10, 32) output: (10, 32, 1)
cat depth:1 input: 3 x (10, 64, 1), (10, 32, 1) output: (10, 224, 1)
squeeze depth:1 input: (10, 224, 1) output: (10, 224)
Linear depth:1 input: (10, 224) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
Linear depth:1 input: (10, 32) output: (10, 16)
relu depth:1 input: (10, 16) output: (10, 16)
Linear depth:1 input: (10, 16) output: (10, 8)
relu depth:1 input: (10, 8) output: (10, 8)
Linear depth:1 input: (10, 8) output: (10, 2)
output-tensor depth:0 (10, 2)